APSTA-GE 2006 Fall 2026

Statistics, Math, and Computing Bootcamp

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1 Course description

This course aims to prepare students for the Applied Statistics for Social Science Research or an equivalent program at NYU. We will cover basic programming using the R language, including data manipulation and graphical displays; some key ideas from calculus, including differentiation and integration; basic matrix algebra, including vector and matrix arithmetic; some core concepts in probability, including random variables, discrete and continuous distributions, and expectations; and a few simple regression examples. We will also discuss the use of modern AI and LLMs in computational statistics.

2 Where and when

The course will run from August 17 through August 28. We will meet on weekdays from 9 a.m. to 12 p.m. at 70 Washington Sq S (Bobst), Room LL150, with a Zoom option for those who can't attend in person. Students are strongly encouraged to attend in person, as this course will be highly interactive.

3 Course prerequisites

Students should be fluent in basic algebra and have seen exponential and logarithmic functions. Some programming experience would be helpful but is not required. Students should be prepared to write many small R programs during the course.

4 Course materials and references

Students are expected to have access to R and RStudio. There are many alternative programming editors, including Positron, VS Code, Zed, and others. Most of these editors are AI-native—they integrate AI agents into the programming workflow. When learning how to read and write code, you should avoid AI assistance and use RStudio instead.

Follow these instructions to download and install the required software. In addition, you may find the following free resources helpful as you continue your studies.

4.1 Programming and Data Visualization

- Hands-On Programming with R, Grolemund (2014)
- R for Data Science, Wickham et al. (2023)
- Data Visualization: A Practical Introduction, Healy (2018)

4.2 Calculus

- YouTube: Essence of Calculus, Sanderson (2018a)
- Calculus Made Easy, Thompson (1980)
- Calculus, Herman et al. (2016)

4.3 Probability

- YouTube: Probability Animations
- YouTube: Statistics 110 @ Harvard
- Introduction to Probability, Blitzstein and Hwang (2019)
- Introduction to Probability Cheat Sheet v2, Chen (2015)

4.4 Statistics

- Regression and Other Stories, Gelman et al. (2020)

4.5 Linear Algebra (for those who need it)

- YouTube: Essence of Linear Algebra, Sanderson (2018b)
- Introduction to Linear Algebra, Boyd and Vandenberghe (2018)
- Matrix Cookbook, Petersen and Pedersen (2012)

5 Tentative schedule

The following tentative schedule assumes ten three-hour sessions, during which we plan to cover the following material.

- **(1) The Big Picture and Introduction to R**

  We will briefly discuss the motivation behind the topics in the class and give some examples of the types of regression problems you may encounter. We will use data on Space Shuttle O-ring damage and John Snow's (the other one) cholera data as motivating examples. We will introduce programming principles in R, including the RStudio IDE, R objects, built-in functions, and writing your own functions.

  - *Hands-On Programming with R*, Part 1.
  - R for Data Science, Introduction.
  - YouTube: Essence of Calculus, Videos 1, 2, 3.

- **(2) Plotting, Exponentials, Logs, and Derivatives**

  We will review linear, exponential, and logarithmic functions and develop some intuition using compound interest. We will explain and code a softmax function using the log-sum-exp trick. Furthermore, we will introduce derivatives geometrically and symbolically. We will cover basic rules for differentiating functions and use noisy measurements of a falling object as a motivating example. We will also practice plotting lines and curves using base `plot` and `ggplot2`.

  - *Hands-On Programming with R*, Part 2.
  - R for Data Science, Data visualization
  - YouTube: Essence of Calculus, Videos 4, 5, 6.

- **(3) Reshaping Data, Loops, and Maps; Introduction to Integration**

  We will practice data wrangling using the `dplyr` and `tidyr` packages. We will cover the intuition behind integration, the approximation of areas, the definite integral, and basic integration rules. We will learn how to perform numerical and symbolic integration (in one dimension) using R.

  - *Hands-On Programming with R*, Part 3.
  - R for Data Science, Data transformations.

- YouTube: Essence of Calculus, Videos 7, 8, 9.

- **(4) Introduction to Probability 1**

  We will review sample spaces and counting and introduce the axiomatic definition of probability. We will discuss how to solve probability problems using simulations and go through a few famous paradoxes. We will make extensive use of R's `sample` and `replicate` functions.

  - R for Data Science, Data Tidying
  - Introduction to Probability, Chapter 1
  - YouTube: Essence of Calculus, Videos 10, 11, 12.

- **(5) Introduction to Probability 2**

  We will introduce conditional probability, the law of total probability, and independence and discuss a few famous paradoxes. We will write R simulations to check analytic solutions.

  - Introduction to Probability, Chapter 2

- **(6) Introduction to Probability 3**

  We will introduce random variables, PDFs, and CDFs. We will cover Bernoulli, binomial, uniform, normal, and exponential RVs. We will discuss the concept of expectation. We will use R's distribution functions to simulate realizations of RVs and compute their properties.

  - Introduction to Probability, Chapter 3

- **(7) Introduction to Linear Algebra**

  We will review the basics of linear algebra, including vector and matrix addition and multiplication. We will practice matrix and vector arithmetic in R. Time permitting, we will solve the overdetermined system of linear equations for the falling-object problem from Lecture 2.

  - YouTube: Essence of Linear Algebra, Videos 1, 2, 3

- **(8) Statistical Inference**

  We will cover basic topics in statistical inference, including sampling distributions, standard errors, classical confidence intervals, bias and uncertainty, and problems associated with the uncritical use of statistical significance in decision-making.

  - Regression and Other Stories, Chapter 4

- **(9) Analysis Workflow and Linear Regression**

  We will introduce an analysis workflow using a dataset containing ratings of red wines. We will use the `rstanarm` package, but the same model can be fit with R's `lm()` function. We will attempt to determine what makes a good wine. After doing some basic EDA, we will fit several regressions and evaluate how well our model performs.

  - Introduction to Probability, Chapter 3

- **(10) Review and Discussion**

  This is a buffer lecture in which we will cover remaining topics from previous days. Time permitting, we will also briefly review the material, answer questions, and discuss the contents of the final assessment.

6 Assessment

This is a pass/fail class with the following grade distribution.

- Class participation: 50%
- Final assessment (take-home): 50%

To earn class participation credit, students must attend all lectures *and* post in the online lecture discussions on Brightspace under the Online Discussion topic. For each lecture, please ask a question about the material, respond to a question asked by another student, or post an observation.

The final assessment will consist of approximately ten questions covering all major topics in the class. The final will be distributed during the last day of class.

7 Academic Integrity

All students are responsible for understanding and complying with the NYU Steinhardt Policies and Academic Integrity.

8 Students with Disabilities Statement

Students with physical or learning disabilities are required to register with the Moses Center for Students with Disabilities at 726 Broadway, 2nd Floor (212-998-4980), and to present a letter from the Center to the instructor at the start of the semester in order to be considered for appropriate accommodations.

9 Mental Health Statement

If you are experiencing undue personal and/or academic stress during the semester that may be interfering with your ability to perform academically, the NYU Wellness Exchange (212-443-9999) offers a range of services to assist and support you. I am available to speak with you about stress related to your work in my course, and I can assist you in connecting with the Wellness Exchange. Additionally, if you anticipate any challenges with completing the assignments, readings, exams, and other work required in this course, I encourage you to register with the Moses Center (212-998-4980) in advance so that you may be granted the proper academic accommodations.

References

Blitzstein, Joseph K., and Jessica Hwang. 2019. *Introduction to Probability*. Second edition. Crc Press/Taylor & Francis Group.

Boyd, Stephen P., and Lieven Vandenberghe. 2018. *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*. Cambridge University Press.

Gelman, Andrew, Jennifer Hill, and Aki Vehtari. 2020. *Regression and Other Stories*. Analytical Methods for Social Research. Cambridge University Press.

Grolemund, Garrett. 2014. *Hands-on Programming with r*. First edition. O'Reilly. https://rstudio-education.github.io/hopr/.

Healy, Kieran. 2018. *Data Visualization: A Practical Introduction*. Princeton University Press.

Herman, Edwin, Gilbert Strang, and OpenStax. 2016. *Calculus Volume 1*. https://d3bxy9euw4e147.cloudfront.net/oscms-prodcms/media/documents/CalculusVolume1-OP.pdf.

Petersen, Kaare Brandt, and Michael Syskind Pedersen. 2012. *The Matrix Cookbook*. https://www.freetechbooks.com/the-matrix-cookbook-t435.html.

Sanderson, Grant. 2018a. *Essence of Calculus*. (YouTube), November 23. https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr.

Sanderson, Grant. 2018b. *Essence of Linear Algebra*. (YouTube), November 23. https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab.

Thompson, Silvanus P. 1980. *Calculus Made Easy: Being a Very-Simplest Introduction to Those Beautiful Methods of Reckoning Which Are Generally Called by the Terrifying Names of the Differential Calculus and the Integral Calculus*. 3d ed. St. Martin's Press.

Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund. 2023. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. 2nd edition. O'Reilly Media.